DAC Testnet

DAC Wallet Intelligence Layer – Function Task Validation & Output Parity Testing

Final / Advanced Phase — Truebit Function-Task Mode Validation

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Environment	WSL Ubuntu · Node.js v24.14.1 · Git v2.43.0 · DAC Testnet Data Snapshot
Report Type	Application Development Report — Function-Task Validation & Output Parity Testing
Version Scope	Wallet Intelligence Layer v2.0.2 · dac_wallet_intelligence task from PR #2
Relates to	DAC Wallet Intelligence Layer Technical Report (28 May 2026)

Community-built engineering validation for DAC Testnet analytics and infrastructure tooling.

Executive Summary

This report documents the final / advanced validation phase of DAC Wallet Intelligence Layer: comparing the local browser output of Wallet Intelligence Layer v2.0.2 against the dac_wallet_intelligence function-task execution from the DAC / Truebit Etherscan API task-library PR #2.

The goal was to confirm whether the user-facing browser implementation can be reproduced by a task-library execution path. The function-task was executed locally with Node.js, using a prepared input derived from the WIL browser raw output. The result confirms successful task execution and strong parity across core metrics and portfolio intelligence, while also exposing scoring model drift in the reputation and Sybil-risk layers.

Component / Finding	Status
WIL browser output captured	**Confirmed**
Function-task executed locally	**Confirmed**
Core metrics parity	**Confirmed**
Portfolio intelligence parity	**Confirmed**
Reputation scoring parity	**Mismatch — model drift**
Advanced WIL modules in function-task	**Out of scope for the current simplified function-task implementation**
Final validation status	**Partial validation complete**

Validation Scope

This validation focuses only on the function-task comparison phase. It does not re-evaluate the full Wallet Intelligence Layer frontend, the Wallet Status SBT deployment, or DAC Sender NFT Launchpad integration. Those were covered in the prior Wallet Intelligence Layer technical report.

Browser output	Wallet Intelligence Layer v2.0.2 raw output
Function-task output	dac_wallet_intelligence task executed locally from PR #2
Comparison target	Field-level parity between browser output and task output
Execution mode	Local Node.js execution — not full Truebit network settlement
Baseline wallet	0x870ad63acc507cdfd878F170606d19ae78988AFE
Overall result	PARTIAL_MATCH_WITH_SCORING_MODEL_DRIFT

Methodology

The validation was performed in four steps:

1. Capture raw output from the Wallet Intelligence Layer v2.0.2 browser interface.
2. Normalize the browser output into comparable core fields.
3. Prepare a function-task input file using the expected dac_wallet_intelligence input shape.
4. Execute the JavaScript task locally and compare output fields against the browser baseline.

Local Execution Commands

```
node -v
# v24.14.1

git --version
# git version 2.43.0

git clone https://github.com/dacblockchain/truebit-etherscan-api-task-library.git
cd truebit-etherscan-api-task-library
git fetch origin pull/2/head:wil-pr-2
git checkout wil-pr-2
cd function-tasks/dac_wallet_intelligence/js/src
node main.js
cat output.txt
```

Execution Evidence

The terminal log confirms that the PR branch was fetched and checked out as wil-pr-2, the task folder contained main.js and config.json, input.txt was created, and node main.js produced output.txt. The captured function-task output is included in the comparison package as function-task-output.json.

Validation Artifacts

Artifact	Purpose
`browser-output.json`	Raw Wallet Intelligence Layer v2.0.2 browser output
`function-task-output.json`	Output produced by local dac_wallet_intelligence task execution
`comparison-result.json`	Machine-readable field comparison result
`validation-report.md`	Human-readable intermediate validation report
`This report`	Formal technical report for the final / advanced phase

Browser Output Baseline

Wallet	0x870ad63acc507cdfd878F170606d19ae78988AFE
App Version	v2.0.2
Status	FULL
Mode	EXPLORER_PRIMARY
Chain	DAC Testnet / 21894
Native Balance	3.99033563 DACC
Total Transactions	1,162
NFT Transfers	213
Total Collections	21
Total NFTs	213
Reputation Score	69
Pattern Health Score	98
Inception Rank	SENTINEL

Function-Task Output Snapshot

Wallet	0x870ad63acc507cdfd878F170606d19ae78988AFE
Native Balance	3.99033563 tDACC
Total Transactions	1,162

NFT Transfers	213
Total Collections	21
Total NFTs	213
Activity Level	HIGH
Portfolio Style	NFT HEAVY
Reputation Score	95
Reputation Level	ELITE
Sybil Risk	LOW

High-Level Result

`PARTIAL_MATCH_WITH_SCORING_MODEL_DRIFT`

Core input metrics and portfolio intelligence mostly match. Reputation and scoring outputs diverge because the browser WIL v2.0.2 implementation uses a more advanced policy with stake, rank, and heuristic layers, while the function-task implementation is a simplified analytics task.

Result	Count
MATCH	**16**
PARTIAL	**2**
MISMATCH	**5**

Field-Level Comparison

Field	Category	Browser Output	Function-Task	Result	Notes
`wallet`	Core identity	0x870ad6...AFE	0x870ad6...AFE	MATCH	Same wallet target.
`native token label`	Core metric	DACC	tDACC	PARTIAL	Label differs: browser uses DACC, task uses tDACC.
`native balance`	Core metric	3.99033563	3.99033563	MATCH	Same numeric balance.
`totalTransactions`	Core metric	1162	1162	MATCH	Transaction count matches.
`nftTransfers`	Core metric	213	213	MATCH	NFT transfer count matches.

`totalCollections`	Core metric	21	21	MATCH	Collection count matches.
`totalNFTs`	Core metric	213	213	MATCH	NFT holding count matches.
`activityLevel`	Classification	VERY HIGH	HIGH	MISMATCH	Threshold / model differs between browser WIL and task.
`engagementType`	Classification	ADVANCED TESTNET USER	ADVANCED TESTNET USER	MATCH	Engagement label matches.
`nftParticipationRatio`	Derived metric	18.33%	0.18	PARTIAL	Browser formats percent string; task returns decimal.
`diversityScore`	Classification	HIGH	HIGH	MATCH	Diversity label matches.
`portfolio totalCollections`	Portfolio	21	21	MATCH	Portfolio collection count matches.
`portfolio totalNFTs`	Portfolio	213	213	MATCH	Portfolio NFT count matches.
`topCollection name`	Portfolio	Nyxia	Nyxia	MATCH	Top collection matches.
`topCollection amount`	Portfolio	60	60	MATCH	Top collection amount matches.
`concentration %`	Portfolio	28.17	28.17	MATCH	Same percentage.
`concentration level`	Portfolio	MEDIUM	MEDIUM	MATCH	Concentration level matches.
`portfolioStyle`	Portfolio	NFT HEAVY	NFT HEAVY	MATCH	Portfolio style matches.
`walletArchetype`	Portfolio	STANDARD USER	STANDARD USER	MATCH	Wallet archetype matches.
`reputationScore`	Reputation	69	95	MISMATCH	Browser uses locked WIL policy with stake, rank, and heuristics; task uses simpler scoring.
`reputationLevel`	Reputation	MEDIUM	ELITE	MISMATCH	Mismatch caused by different reputation score.
`trustProfile`	Reputation	VERIFIED INCEPTION PARTICIPANT	ADVANCED TESTNET PARTICIPANT	MISMATCH	Mismatch caused by task scoring model difference.
`sybilRisk`	Reputation	HIGH	LOW	MISMATCH	Browser reputation-level sybilRisk differs from task output.

Technical Interpretation

The function-task successfully reproduces the wallet identity, numeric input metrics, NFT totals, portfolio concentration, top collection, portfolio style, and basic engagement type. This confirms that the task-library execution path can process the same wallet profile input and generate usable analytics output.

The mismatches appear primarily in the reputation and Sybil-risk layer. This is expected because the current task implementation is simpler than the browser WIL v2.0.2 implementation. The browser version includes stake-flow classification, DAC Inception Rank signal, a locked WIL scoring policy, explorer-only Sybil heuristics, historical activity windows, and Dynamic Intelligence Badge state.

```
Core metric parity          : confirmed
Portfolio parity            : confirmed
Basic task execution        : confirmed
Advanced WIL parity         : out of scope for the current simplified function-task
implementation
Scoring policy alignment    : required
```

Missing or Out-of-Scope in Current Function-Task

- stakedDacc / stake-flow classifier
- DAC Inception Rank score and inferred rank
- WIL-v1.5.2 scoring policy breakdown
- EOH-v1.5.2 explorer-only Sybil heuristics
- Historical activity windows
- dynamicIntelligenceBadge / Wallet Status SBT state
- knownCollectionRegistry matching
- Raw NFT collection knownCollection metadata

Final Validation Status

Component / Finding	Status
WIL frontend output captured	**YES**
Function-task execution completed	**YES**
Core metrics comparison	**MATCH**
Portfolio comparison	**MATCH**
Reputation comparison	**MISMATCH**
Advanced WIL modules comparison	**Out of scope for current simplified function-task**
Final status	**PARTIAL VALIDATION COMPLETE**

Recommendations

For Function-Task Parity

1. Upgrade the dac_wallet_intelligence task to support the locked WIL-v1.5.2 scoring policy.
2. Add DACC stake-flow input handling and score calculation.
3. Add DAC Inception Rank signal handling and inferred rank output.
4. Add explorer-only Sybil heuristic components and pattern health scoring.
5. Add historical activity window support for 7D, 30D, and all-time views.
6. Add dynamic badge-compatible normalized output fields, without requiring SVG rendering in the task environment.

For Documentation

1. Document the current task as a simplified analytics execution mode.
2. State clearly that full WIL v2.0.2 parity is out of scope for the current simplified function-task implementation in the task-library version.
3. Use this report as the baseline for future parity testing after the task is upgraded.

Conclusion

The final / advanced phase has been completed as a partial validation baseline. The dac_wallet_intelligence function-task executed successfully and reproduced the core wallet metrics and portfolio intelligence output from the Wallet Intelligence Layer browser implementation. This confirms that the task-library path can serve as a reproducible execution mode for foundational wallet analytics.

However, full parity with WIL v2.0.2 has not yet been reached. The current function-task does not include the advanced scoring, stake, rank, historical activity, Sybil heuristic, and dynamic badge modules used by the browser implementation. The correct status is therefore partial validation complete, with scoring policy alignment required for full parity.

Final Phase	Truebit Function-Task Mode
Execution Status	Completed
Validation Status	Partial validation complete
Primary Finding	Core metrics and portfolio output match; scoring layer diverges.
Next Step	Upgrade function-task to match WIL v2.0.2 advanced policy layers.

Appendix A — Function-Task Output


```
{
  "wallet": "0x870ad63acc507cdfd878F170606d19ae78988AFE",
  "nativeBalance": {
    "token": "tDACC",
    "balance": 3.99033563
  },
  "metrics": {
    "totalTransactions": 1162,
    "nftTransfers": 213,
    "totalCollections": 21,
```

```
    "totalNFTs": 213
  },
  "activityAnalytics": {
    "activityLevel": "HIGH",
    "engagementType": "ADVANCED TESTNET USER",
    "nftParticipationRatio": 0.18,
    "diversityScore": "HIGH"
  },
  "portfolioIntelligence": {
    "totalCollections": 21,
    "totalNFTs": 213,
    "topCollection": {
      "collection": "Nyxia",
      "symbol": "NXIA",
      "contract": "0x7cdd0e631372747764b70b9d97948932c1ebc706",
      "amount": 60
    },
    "concentration": {
      "percentage": 28.17,
      "level": "MEDIUM"
    },
    "portfolioStyle": "NFT HEAVY",
    "walletArchetype": "STANDARD USER"
  },
  "reputationScoring": {
    "reputationScore": 95,
    "reputationLevel": "ELITE",
    "trustProfile": "ADVANCED TESTNET PARTICIPANT",
    "sybilRisk": "LOW"
  }
}
```